

Intergenerational Housing Misallocation and Fertility

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Housing and Fertility

Children increase how big of a house you need. This has implications for:

- Renting: large rental units are scarce.
- Owning: a family-sized home requires liquid wealth and a down payment.
- Wealth: older households retain much of the large owner stock after children leave.

Physical space is key. How much does it affect fertility?

What kind of housing policy can improve access to family-sized homes?

→ focus on space constraints and tenure choice

This Paper

Need a structural model where fertility, tenure, housing size, and saving are joint.

Theory: a simple model of access to family-sized housing

- Rental limits and down payments raise the shadow value of space.
- Retention by older owners can misallocate the existing housing stock.

Quantification: a lifecycle model with income risk, tenure choice, transaction costs, bequests, and entry

Goal: evaluate housing policies that change access when households make fertility decisions.

Environment

Young households

- Draw income and liquid wealth, $i = (y_i, b_i)$.
- Choose entry, tenure, housing, completed fertility, and saving.
- Become old owners or old renters next period.

One stock $\bar{H}$ clears at rental user cost $r = \rho P$. Renting is capped at $\bar{h}^R$ and owning at $\bar{h}^O$, with $\bar{h}^R < \bar{h}^O$.

Old households

- Owners choose how much housing to retain and the estate to leave.
- Renters demand housing without an ownership lock-in wedge.
- All households exit after old age.

Preferences

A young household receives utility from adult consumption, residual space, and completed fertility:

$$u_i^Y = \log(c_i - \chi n_i) + \alpha \log(h_i - \kappa n_i) + \vartheta \log n_i.$$

- Each child requires χ units of consumption and κ units of housing.
- Renting and owning have the same market user cost per unit of space.
- Tenure matters because only ownership reaches the largest units and buying requires equity.

A buyer must satisfy $(1 - \phi)Ph_i \leq b_i$: current liquid wealth, not lifetime income, governs access to owner housing.

Housing, Tenure, and the Effective Cap

Using $P = r/\rho$, the maximum housing available in each tenure is

$$H_i^R = \bar{h}^R, \quad H_i^O = \min \left\{ \bar{h}^O, \frac{b_i \rho}{(1 - \phi)r} \right\}.$$

- Renters are constrained by the largest available rental unit.
- Owners may be constrained by the physical menu or by the down payment.
- A household can afford the flow cost of space yet remain unable to purchase it.

The relevant friction is therefore not tenure by itself, but the household-specific cap on family-sized space.

The Access Gap

For a constrained household, the marginal value of space exceeds its effective price by $\zeta_i^m \geq 0$:

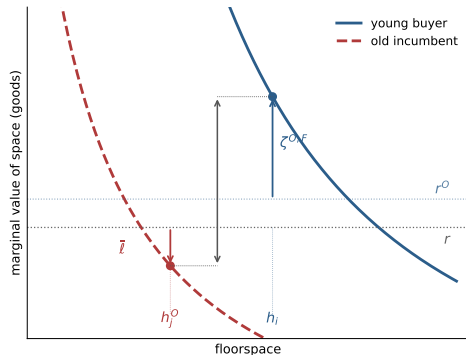
$$\frac{\alpha(c_i^m - \chi n_i^m)}{h_i^m - \kappa n_i^m} = r^m + \zeta_i^m.$$

The fertility first-order condition becomes

$$\underbrace{\frac{\vartheta(c_i^m - \chi n_i^m)}{n_i^m}}_{\text{willingness to pay for a child}} = \underbrace{\chi + \kappa(r^m + \zeta_i^m)}_{\text{full price of a child}}.$$

Interpretation: limited access to space acts like a household-specific tax $\kappa\zeta_i^m$ on children, even when market prices do not change.

Intergenerational Misallocation

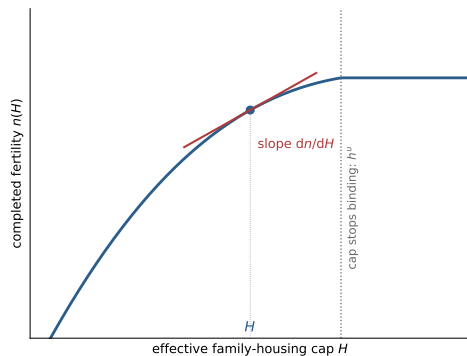


Moving one unit from a locked-in old incumbent to a financing-constrained young buyer produces surplus

$$\zeta_i^{O,F} + q\bar{\ell} + \bar{\ell} > 0.$$

- The buyer values space above effective user cost.
- The incumbent values retained space below the market rent because selling realizes a taxable gain.
- The common market price cancels.

Fertility and Policy



Along a binding cap, more space has two effects:

- it makes the child's housing requirement easier to meet;
- it tightens the budget through the cost of additional space.

Fertility rises when the housing channel dominates:

$$\chi \leq (1 + \Lambda) \frac{\kappa r^m}{\alpha}.$$

Exposure and pass-through determine the aggregate effect of a housing policy.

Quantitative Lifecycle Model

18	18–42	66	85
entry	fertile window	retirement	maximum age

- Households face persistent earnings risk and save in liquid assets.
- A childless household makes a one-time discrete completed-family-size choice; children then age and leave home.
- Each period households choose renting or one of five owner sizes and adjust saving.
- Entry and the stationary population scale respond in counterfactuals.

The model preserves the analytical mechanism while adding the lifecycle margins needed for measurement and policy.

Preferences and Bequests

Dependent children shift minimum consumption and housing requirements:

$$\bar{c}(n, s) = \bar{c}_0 + \bar{c}_n n_s, \quad \bar{h}(n, s) = \bar{h}_0 + \bar{h}_{\text{jump}} \mathbf{1}\{n_s \geq 1\} + \bar{h}_n n_s.$$

Flow utility is CRRA over residual consumption and usable housing services:

$$u = \frac{[(c - \bar{c})^{\alpha_c} \mathbf{h}^{1-\alpha_c}]^{1-\sigma}}{1-\sigma} + \psi_{\text{child}} n_s,$$

where owners receive service premium χ_O after household needs are removed.

At death, households value liquid wealth plus the gross value of the house through a warm-glow bequest motive.

Housing, Tenure, and Budget Constraints

Renters

- Choose rooms continuously up to $\bar{h}^R = 6$.
- Cannot borrow: $b' \geq 0$.
- Pay rental user cost r .

Owners

- Choose $h \in \{2, 4, 6, 8, 10\}$ rooms.
- Can borrow up to $\phi = 0.80$ of house value.
- Pay depreciation, property tax, and a 6% sale cost when adjusting.

Housing supply is upward sloping,

$$H^S(r) = \bar{H} \left(\frac{r}{\bar{r}} \right)^\eta, \quad \eta = 1.75,$$

so higher demand is absorbed by both quantities and prices.

Family Size, Housing, and Equilibrium

A childless household in the fertile window chooses family size first, then its current housing position:

$$n' \in \{0, 1, 2\}, \quad h' \in \{0, 2, 4, 6, 8, 10\}.$$

- $h' = 0$ means renting; positive values are owner rungs.
- A positive family-size choice fixes completed fertility and activates dependent-child needs immediately.
- Type-I extreme-value shocks smooth both family-size and tenure choices.
- The equilibrium combines optimal policies, the stationary distribution, entry and population scale, and housing-market clearing.

A birth can therefore change tenure and occupied space in the same period, rather than only affecting future housing demand.

Calibration

Assigned outside estimation

- Lifecycle timing and survival
- $\sigma = 2$, real return, taxes, depreciation
- $\phi = 0.80$ and sale cost 6%
- Income process and housing menus
- Supply elasticity and entrant wealth

Estimated internally

- Patience and consumption weight
- Consumption and housing needs
- Value and dispersion of family size
- Ownership premium and tenure dispersion
- Housing-supply scale and bequests

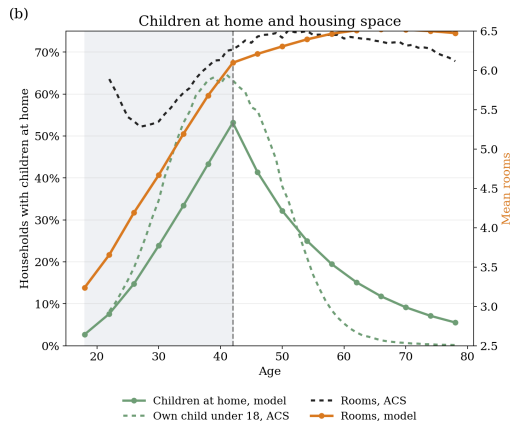
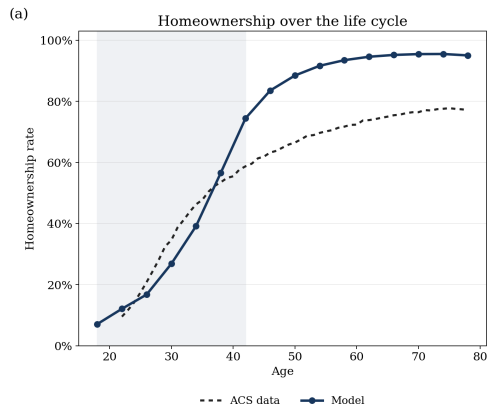
Fourteen parameters are jointly disciplined by CPS fertility, ACS housing and tenure, and PSID wealth and estate moments.

Model Fit

Moment	Data	Model
Completed fertility / childlessness	1.918 / 0.188	2.034 / 0.189
First-child rooms increment	0.664	0.681
Family ownership gap	0.168	0.219
Ownership, ages 30–55	0.575	0.658
Ownership, ages 25–34	0.341	0.274
Ownership, ages 65–75	0.764	0.954
Renter rooms / owner–renter rooms gap	3.805 / 2.419	3.963 / 2.323
Childless-owner share with ≥ 6 rooms	0.596	0.760
Young liquid wealth / income	0.179	0.328
Estate median / income	6.501	6.431
Older nonhousing-wealth share	0.608	0.610

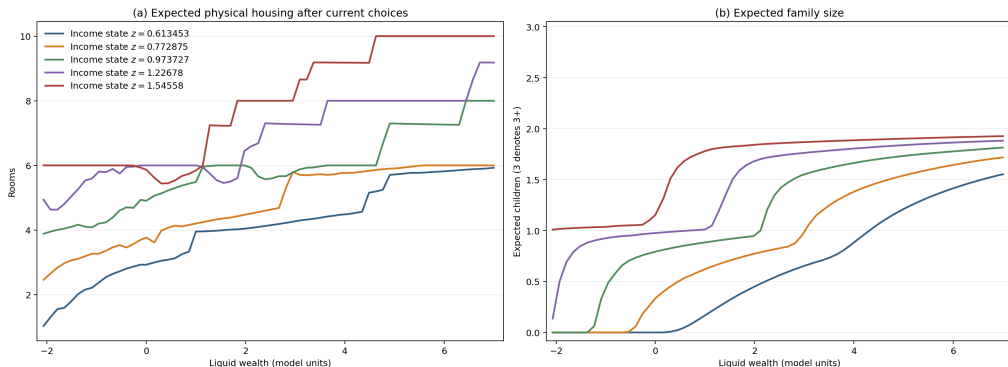
The largest tension is excessive persistence in homeownership: young households buy too late, while older households rarely exit ownership.

Lifecycle Housing Patterns



Homeownership is understated through the mid-thirties and overstated from the early forties onward. The child-at-home profile has the right hump but is delayed and too persistent.

Housing and Family Size at Age 42



At age 42, higher-income households choose children and larger homes at lower liquid-wealth levels. The discrete owner ladder generates sharp changes in physical rooms.

Quantitative Evidence on the Mechanism

Among fertile renters whose positive-family branch reaches the rental cap:

$$\underbrace{r}_{0.130} + \underbrace{\zeta_i^R}_{0.178} = \underbrace{0.307}_{\text{marginal value of another room}} = 2.37 r.$$

Holder of owner homes with ≥ 6 rooms	ACS	Model
Households age 46+	71.8%	80.7%
Households with no children at home	54.9%	74.8%

The households facing the highest marginal value of family space are distinct from those holding most of the family-sized owner stock.

Housing Policy

Policy	Δ total births	Δ housing price
Purchase grant for family-sized homes	+2.1%	+0.90%
Higher property tax plus purchase grant	+2.7%	-18.44%

- The grant covers roughly one-third of the required down payment on an eligible home.
- The package also raises the annual property-tax rate on owner housing from 1 to 2 percent.
- Entry, population, and the housing price adjust in both exercises.

Conclusion

- A model of completed fertility with tenure choice, saving, and a lifecycle of housing need.
- Children raise housing need; rental limits and down payments make space especially costly for young households.
- Older households hold most large owner homes. With retention wedges, a cleared market need not allocate them efficiently.
- Aim is policy: purchase assistance raises births, and the tax-plus-grant package changes both access and prices.

More work is needed: sequential fertility, income risk, old-age downsizing, and across-market relocation.